

Technical Configuration Reference Guide

Document Reference: CAW-TECH-CFG-2026-01
System: Case Ace v2.0 Client Consultation Recording & Note Drafting System
Organisation: Citizens Advice Wandsworth (CAW)
Status: Formally Approved Technical Reference
Classification: Internal / Technical Reference

1. Environment Variables Reference

ENVIRONMENT VARIABLES SPECIFICATION		
Variable Name	Default Value	Description
<code>`NODE_ENV`</code>	<code>`production`</code>	Application execution environment
<code>`PORT`</code>	<code>`8080`</code>	Backend HTTP port (injected by Cloud Run)
<code>`APP_ENV`</code>	<code>`pilot`</code>	Selects the config block in backend/src/config
<code>`CORS_ORIGIN`</code>	<code>`https://caseace.adviceintelligence.tech`</code>	Allowed client web origin
<code>`GCP_PROJECT_ID`</code>	<code>`case-ace-v2`</code>	Google Cloud Project identifier
<code>`GCP_REGION`</code>	<code>`europe-west2`</code>	Pinned UK sovereign GCP region
<code>`GCP_SPEECH_RECOGNIZER`</code>	<code>`chirp2-uk-english`</code>	Google STT v2 Recognizer resource
<code>`GCP_VERTEX_MODEL`</code>	<code>`gemini-1.5-pro-002`</code>	Large Language Model for drafting
<code>`GCP_VERTEX_DRAFT_TEMPERATURE`</code>	<code>`0.2`</code>	Low temperature for factual output
<code>`SESSION_IDLE_TIMEOUT_SECONDS`</code>	<code>`900`</code>	Inactivity timeout (15 minutes)
<code>`AUDIT_LOG_RETENTION_DAYS`</code>	<code>`365`</code>	Retention period for non-PII logs
<code>`WEBEX_CLIENT_ID`</code>	<code>*(Encrypted Secret)*</code>	Cisco Webex OAuth 2.0 Client ID
<code>`WEBEX_REDIRECT_URI`</code>	<code>`https://caseace.adviceintelligence.tech/api/auth/webex/callback`</code>	Webex OAuth callback URI
<code>`ENTRA_REDIRECT_URI`</code>	<code>`https://caseace.adviceintelligence.tech/auth/callback`</code>	Microsoft Entra ID callback URI
<code>`ALLOW_TOTP_IN_PILOT`</code>	<code>`true`</code>	Interim TOTP login while Entra ID is unavailable
<code>`JWT_SECRET`</code>	<code>*(Secret Manager: case-ace-jwt-secret)*</code>	Session token signing key. Never use the committed fallback
<code>`VITE_APP_ENV`</code>	<code>`pilot`</code>	Build-time only. Omitting it makes the SPA fail closed to <code>`local`</code>

2. Client-Side Audio & Buffer Parameters

```
export const AUDIO_CONFIG = {
  sampleRate: 16000, // 16 kHz mono audio (optimal for Whisper & Chirp 2)
  channelCount: 1, // Mono recording channel
  maxDurationSeconds: 3600, // 60-minute hard session limit
  chunkSizeFrames: 4096, // Web Audio API ScriptProcessor / AudioWorklet buffer
  quantizationBits: 16, // 16-bit PCM Linear WAV
};
```

3. Local Whisper WASM Engine Parameters

```
export const WHISPER_CONFIG = {
  modelName: 'whisper-base-en-v3', // 74M parameter quantized model
  threads: 4, // WebAssembly SIMD worker threads
  temperature: 0.0, // Greedy decoding for maximum deterministic output
  language: 'en', // Enforce English language transcription
  beamSize: 5, // Beam search width
};
```

4. Prompt & Model Version Registry